Aouad, Lykouris & Zhong (2026)

Human-AI Productivity Paradoxes — Section 2

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github.com/alexanderquispe/ai-01-aouad

Question. When does AI assistance make a worker *less* productive?

The single mechanism. AI is a *perfectly substitutable input*: skill, effort and assistance enter production only through their sum.

$$x \;=\; s + e + a$$

The agent's problem. Choose effort e to maximise output net of a *linear* cost:

$$\max_{e \geq 0} \; p(s + e + a) \;-\; \gamma e$$

- p weakly increasing, concave, twice differentiable; $\gamma > 0$.
- Only constraint: $e \geq 0$ — and it binds. That is the whole result.

Define x^* as the *largest* maximiser of $p(x) - \gamma x$:

$$x^* \;=\; \max \arg\max_{x} \left[\, p(x) - \gamma x \,\right]$$

Regularity condition (without it x^* need not exist):

$$\limsup_{x \to \infty} \frac{p(x)}{x} \;<\; \gamma$$

Then

$$e^*(s,a) = \left(x^* - s - a\right)_+, \qquad p^*(s,a) = \max\{\, p(x^*),\; p(s+a)\,\}$$

Intuition. The agent has one target level of total input; he tops it up with effort, and once skill plus AI already reach it he simply stops working.

Reproduced Proposition 2.1 by hand — the interior-versus-corner split, with no differentiation in the proof (`hand/`).

Assumptions I considered relaxing

- **Linear cost → convex.** *Dead end*: the authors already do it in Appendix D (Props. D.3, D.6). Replication, not extension.
- **Myopia.** A two-period agent who internalises skill dynamics. Never relaxed anywhere in the paper — this is the live one.
- **Complementarity inside** $p(\cdot)$ rather than as an added interaction term. Harder: §4.4 and §5.3 partly pre-empt it.

Also produced a full step-by-step tutorial of the paper and two lecture decks (`extra/`).

What I asked. "What is the most natural extension of this model?"

What the model answered. Relax the linear cost $c(e) = \gamma e$ to a strictly convex $c(e)$ — presented confidently as an open direction.

My verdict: incorrect as an extension.

- Appendix D is titled *"Extension to convex cost functions"* and redoes Sections 2–4 under strictly convex cost.
- So the proposal is already in the paper. I checked the appendix directly rather than taking the answer at face value.
- *Caveat worth keeping:* Appendix D covers §§2–4 but **not** §5, so convex cost applied to the skill-polarisation result is genuinely untouched.

The step I re-derived by hand is in `hand/` — I did not trust the algebra of the first-order condition until I had written it out.